

Deep Learning-based Control of Multi-Port Solid State Transformer

Naheel Faisal Kamal¹, Abdullah Berkay Bayindir², Ali Sharida¹, Sertac Bayhan^{1,3}, Haitham Abu-Rub^{1,4}

¹Qatar Environment and Energy Research Institute (QEERI), Hamad Bin Khalifa University, Doha, Qatar

²Department of Electrical & Computer Engineering, Texas A&M University, College Station, TX, USA

³Department of Electrical & Electronic Engineering, Gazi University, Ankara, Turkey

⁴College of Science and Engineering (CSE), Hamad Bin Khalifa University, Doha, Qatar

nkamal@hbku.edu.qa

Abstract—This paper proposes a deep learning-based control strategy for a multi-port solid-state transformer (SST) to achieve accurate and efficient power flow regulation among its ports. A three-port SST architecture is developed along with its power electronic bridges. Then, the phase-shift angles between the bridges are systematically swept to span the full operating range of power transfer among the ports. The resulting dataset is used to train a deep neural network (DNN) that learns the nonlinear relationship between power flow and phase-shift modulation while taking into consideration different voltage levels, and passive parameters range. Once trained, the DNN directly maps the power reference signals to optimal phase-shift commands that minimize the tracking error between the measured and reference power. Moreover, the proposed approach eliminates the need for iterative optimization or analytical power flow models. This enables overcoming the challenges associated with the highly coupled and nonlinear dynamics of multi-port SSTs. In addition, the method significantly reduces the online computational burden while maintaining fast dynamic response and high control accuracy. Experimental results validate the effectiveness of the proposed deep learning-based controller under various operating conditions and power transfer scenarios.

Index Terms—Multi-port transformer, multi active bridge, data-driven control, deep neural network.

I. INTRODUCTION

The increasing penetration of renewable energy sources, distributed energy storage systems, and electric mobility technologies accelerated the development of advanced multi-port power conversion systems [1]. These systems enable bidirectional power exchange among sources and loads operating at different voltage and power levels. Among the various converter topologies proposed for such applications, the Multi-Active Bridge (MAB) converter emerged as a promising solution due to its ability to interface multiple ports through a common high-frequency transformer (HFT) [2]. The MAB converter offers several advantages, including high efficiency, bidirectional power flow, galvanic isolation, high power density, and flexible power sharing among ports [3]. These features make the MAB converter particularly attractive for solid-state transformers, hybrid energy storage systems, DC microgrids, and renewable-integrated charging infrastructures. However, accurate and robust power flow control in MAB converters is a key challenge due to their strong nonlinear

behavior, tight coupling among ports, and sensitivity to parameter variations [4].

A wide range of control strategies were proposed in the literature to address the complex dynamics of MAB converters [5]. Conventional decentralized linear controllers, such as Proportional-Integral (PI) [6] controllers, are commonly employed due to their simplicity and ease of implementation. However, they fail to capture the inherently nonlinear power-phase shift relationship of MAB converters. Several nonlinear control techniques were proposed to overcome the uncertainty and nonlinear power-phase shift relationship challenges. Sliding Mode Control (SMC) [7] provides robustness against parameter uncertainties and external disturbances. However, the chattering phenomenon associated with SMC leads to increased switching losses and device stress. Feedback Linearization (FL) [8] was proposed to explicitly cancel system nonlinearities. However, FL relies heavily on accurate system modeling and precise knowledge of parameters. Model Predictive Control (MPC) gained significant attention in power electronics due to its ability to handle multi-variable systems and explicitly incorporate constraints [9]. Despite its strong performance, MPC suffers from high computational complexity, especially in multi-port systems where the number of candidate control combinations increases rapidly with the number of bridges.

Recently, intelligent and data-driven control techniques emerged as a promising alternative for controlling MAB converters [10]. These approaches leverage advances in machine learning to learn the complex nonlinear mapping between control variables and power flow directly from data, without relying on explicit analytical models. Techniques such as fuzzy logic control [11], neural networks [10], and reinforcement learning [12] demonstrated the ability to handle nonlinearities, coupling effects, and operating point variations. Unlike model-based methods, data-driven controllers can inherently deal with unmodeled dynamics and model uncertainty present in real systems. However, many existing intelligent control approaches focused on supervisory or tuning layers, require online optimization or iterative learning, or lack a direct and computationally efficient phase-power mapping suitable for real-time implementation.

This paper proposes a deep learning-based power flow

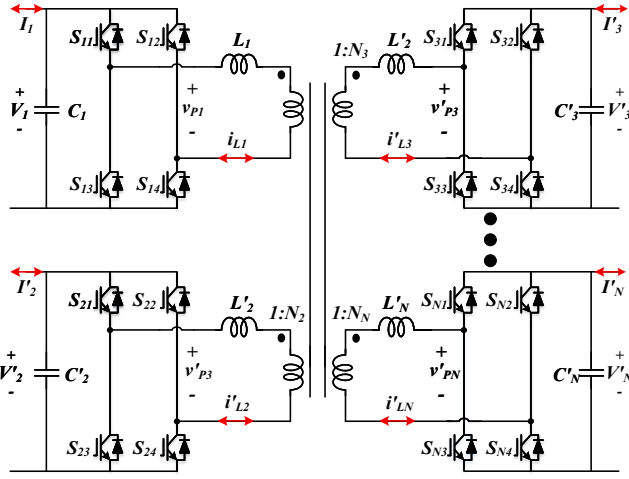


Fig. 1. Circuit scheme of a multi-port SST.

control strategy for a multi-port solid-state transformer based on the MAB converter. The proposed approach employs offline data generation by sweeping the phase-shift angles among converter bridges to cover the full operating range of power transfer. A deep neural network (DNN) is trained to learn the inverse nonlinear relationship between power references and optimal phase-shift commands. Once trained, the DNN directly generates the optimal phase shifts that minimize power tracking errors, thereby eliminating the need for real-time optimization, exhaustive search, or explicit system modeling. As a result, the proposed method significantly reduces online computational burden while effectively handling nonlinearities and coupling effects inherent in multi-port converters.

II. SYSTEM MODEL

A MAB converter consists of multiple full-bridge converter modules, each interfacing a separate DC port. For an N -port MAB converter, N full-bridge converters are connected to an HFT through their dedicated windings, as shown in Fig. 1. Within the MAB converter, the voltage sources are represented by V_1, V_2', V_3' to V_N' for ports 1, 2, 3, to N . Moreover, filtering is provided by C_1, C_2' and C_3' on the first, second and third ports respectively, and by C_N' on the N^{th} port. Each port features a leakage inductance ($L_1, L_2', L_3', \dots, L_N'$), while the switches of H-bridges are represented by S_{ij} , where $i \in [1, 2, \dots, N]$ and $j \in [1, 2, 3, 4]$. The secondary windings of each port are characterized by their turn ratios $1 : N_2, 1 : N_3$ and $1 : N_N$. Lastly, instantaneous voltages ($v_{p1}, v_{p2}', v_{p3}', \dots, v_{pN}'$) and inductor currents ($i_{L1}, i_{L2}', i_{L3}', \dots, i_{LN}'$) are indicated for each port.

To facilitate a simpler analysis of the MAB converter, each full-bridge converter can be considered as an ideal square-wave source, and all ports can be reflected to the primary port. This reflection can be achieved using the following relationships [13].

$$v_{px} = \frac{v'_{px}}{N_x}, \quad i_{Lx} = N_x i'_{Lx}, \quad L_x = \frac{L'_x}{(N_x)^2} \quad (1)$$

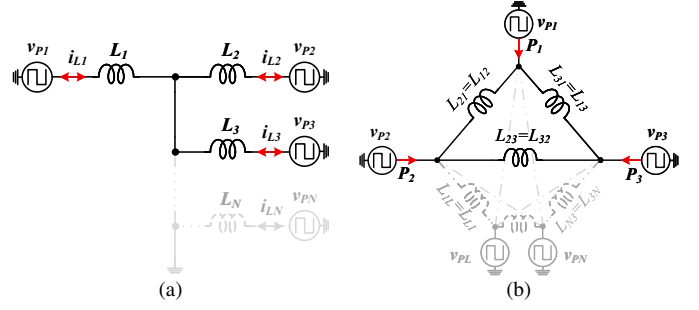


Fig. 2. Equivalent circuits of the MAB converter. (a) Y-type equivalent. (b) Δ -type equivalent.

where x is the port number. As a result, the Y-type equivalent circuit illustrated in Fig. 2a can be achieved. In the Y-equivalent model, all port currents and voltages are represented with respect to the primary side. Such a representation simplifies the examination of the influence of phase shifts and modulation patterns among the ports. For further insight into port interactions and power flow analysis within the system, the Y-type can be transformed into Δ -type equivalent. This transformation can be achieved by calculating the effective inductance between each two ports as follows,

$$L_{xy} = L_x + L_y + L_x L_y \left(\sum_{k \neq x, y}^N \frac{1}{L_k} \right), \quad \forall x \neq y \quad (2)$$

where x and y are sending and receiving ports, respectively. Moreover, the number of effective inductances in a MAB converter can be calculated by the binomial coefficient $C(N, 2)$. Finally, the Δ -type equivalent circuit, illustrated in Fig. 2b, can be used to analyze direct port-to-port power transfer within the MAB converter. In Δ -type MAB circuit, the transferred power P_{xy} between ports x and y based on dual phase shift modulation can be expressed as [14],

$$P_{xy} = \frac{V_{px} V_{py}}{2\pi^2 f_{sw} L_{xy}} \phi_{xy} (\pi - \phi_{xy}), \quad \phi_{xy} = \phi_x - \phi_y \quad (3)$$

where V_{px} and V_{py} are the square-wave amplitudes generated by the H-bridge converters referred to the primary side, L_{xy} is the effective inductance between ports, f_{sw} is the switching frequency, and ϕ_x, ϕ_y denote the phase shift angles of each port, respectively. The relative phase shift (ϕ_{xy}) between ports governs both the direction and the quantity of power transferred. As a result, the total power received/injected by a port can be formalized as,

$$P_x = \sum_{y \neq x}^N P_{xy} = \frac{V_{px}}{2\pi^2 f_{sw}} \sum_{y \neq x}^N \frac{V_{py}}{L_{xy}} \phi_{xy} (\pi - \phi_{xy}) \quad (4)$$

Employing (4) for three ports as a case study of the MAB converter, the net power exchanged in triple active bridge (TAB) converter can be derived as follows,

$$P_1 = \frac{V_{p1} V_{p2} L_3 \phi_{21} (\pi - |\phi_{21}|)}{2\pi^2 f_{sw} (L_1 L_2 + L_2 L_3 + L_3 L_1)} \dots + \frac{V_{p1} V_{p3} L_2 \phi_{31} (\pi - |\phi_{31}|)}{2\pi^2 f_{sw} (L_1 L_2 + L_2 L_3 + L_3 L_1)} \quad (5)$$

$$P_2 = \frac{V_{p1}V_{p2}L_3\phi_{21}(|\phi_{21}| - \pi)}{2\pi^2 f_{sw}(L_1L_2 + L_2L_3 + L_3L_1)} \dots + \frac{V_{p2}V_{p3}L_1(\phi_{31} - \phi_{21})(\pi - |\phi_{31} - \phi_{21}|)}{2\pi^2 f_{sw}(L_1L_2 + L_2L_3 + L_3L_1)} \quad (6)$$

$$P_3 = \frac{V_{p1}V_{p3}L_2\phi_{31}(|\phi_{31}| - \pi)}{2\pi^2 f_{sw}(L_1L_2 + L_2L_3 + L_3L_1)} \dots + \frac{V_{p2}V_{p3}L_1(\phi_{31} - \phi_{21})(|\phi_{31} - \phi_{21}| - \pi)}{2\pi^2 f_{sw}(L_1L_2 + L_2L_3 + L_3L_1)} \quad (7)$$

III. DNN-BASED CONTROL MODEL FOR THE TRIPLE ACTIVE BRIDGE

The proposed data-driven control model is developed to generate the required phase shift commands for a TAB converter. The main objective of this approach is to enable accurate power flow regulation among its three ports. This is achieved by computing the phase shifts of ports 2 and 3 relative to port 1, based on the measured port voltages and the desired power references. In the adopted modulation scheme, the phase shift of port 1 is fixed to zero, and power balance is inherently enforced by the power balance formula,

$$P_1^* = -(P_2^* + P_3^*). \quad (8)$$

Accordingly, the learning problem is reduced to a well-posed inverse mapping between the remaining independent variables as follows,

$$\hat{\phi}_2, \hat{\phi}_3 = \text{DNN}(V_1, V_2, V_3, P_1^*, P_2^*) \quad (9)$$

To close the loop and ensure the power error is considered, the outputs of the DNN are passed to a proportional gain block,

$$\hat{\phi}_{i,t} = \hat{\phi}_i + K_p(P_i^* - P_i) \quad (10)$$

Where $i \in 2, 3$ and the resulting $\hat{\phi}_{i,t}$ are the tuned values of $\hat{\phi}_i$. The block diagram of the overall DNN-based control algorithm is illustrated in Fig. 3.

A. Data Acquisition and Preprocessing

The dataset used to train the DNN-based controller was obtained experimentally from the TAB converter operating under steady-state and dynamic conditions. The data collection process was designed to ensure sufficient variations of system inputs across a wide operating range, including variations in port voltages and bidirectional power flow conditions. During data acquisition, the phase shifts ϕ_2 and ϕ_3 were independently swept over the full control range. The resulting generated dataset is around 380 thousand rows.

Prior to training, the dataset was preprocessed to improve numerical conditioning and learning efficiency. All voltage and power variables were normalized using min-max normalization based on their respective operating ranges. Similarly, the phase shift targets were normalized with respect to their maximum allowable value. This normalization prevents bias due to different magnitudes of inputs and outputs, and it improves the convergence while training.

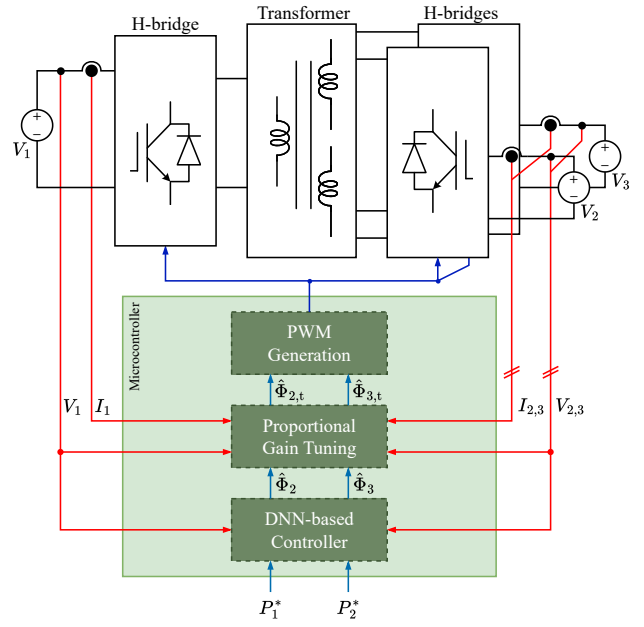


Fig. 3. Control block diagram of the proposed controller.

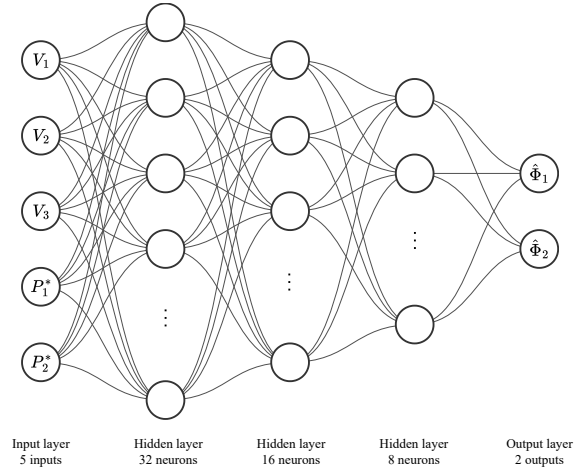


Fig. 4. Architecture of the proposed DNN used for TAB phase-shift generation.

B. Neural Network Architecture and Training

DNN was selected to model the nonlinear inverse relationship between the TAB operating conditions and the required phase shifts. Due to the strong coupling between port voltages, power references, and phase shifts, shallow models are insufficient in capturing the underlying nonlinearities. In contrast, DNNs provide enhanced representational capacity, and enables accurate approximation of the inverse power-flow characteristics.

Several network configurations were evaluated, and a compact architecture was selected to balance performance and computational efficiency. The final model consists of three hidden layers with 32, 16, and 8 neurons. Each neuron employs rectified linear unit (ReLU) activation functions to

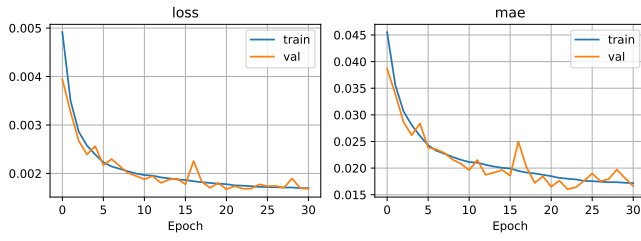


Fig. 5. Training history of the developed DNN.

handle nonlinearities. The output layer uses a linear activation to avoid saturation. The adopted architecture is illustrated in Fig. 4.

The network was trained using the mean squared error loss function and the Adam optimizer. Early stopping was employed to mitigate overfitting and ensure reliable generalization across operating conditions. Fig. 5 shows the training history, in which both the loss and the mean absolute error (MAE) for the training and validation sets converge consistently. The validation set MAE reached 0.0166 after training for 30 epochs, considering the scaled dataset.

C. Model Testing and Validation

Before training, the dataset was split into 70% for training and 30% for testing, where 20% of the training set was used for validation during training. Evaluating the model using the test set after training resulted into a MAE of 0.0167, match the validation error. This shows that the model is robust against data it was not trained with.

To further ensure the effectiveness of the model, Fig. 6 is presented as a use case of Φ_2 and Φ_3 being swept between -45° to 45° with a step of 15° and P_1 – P_3 were originally collected from the measured data. The powers were then fed into the DNN model as reference P_1^* – P_3^* along with the voltages and the predicted phase shifts $\hat{\Phi}_2$ and $\hat{\Phi}_3$ were collected as the results of the DNN's inference. The results match the values of Φ_2 and Φ_3 as specified in the original dataset. To ensure that the predicted $\hat{\Phi}_2$ and $\hat{\Phi}_3$ can achieve the required powers at the ports, the DNN outputs are passed back to the forward mathematical model in (5)–(7) to calculate $P_{1,f}$ – $P_{3,f}$. This validation is useful for the case shown in Fig. 7 where the model predicts phase shifts for smooth synthetic changes in the reference power where $P_{1,f}$ – $P_{3,f}$ follow the same shape of P_1 – P_3 .

D. Implementation and Deployment

The proposed DNN controller model was developed using TensorFlow and Keras as the training environment. To enable a C-compatible real-time execution on the STM32H745 microcontroller, the trained model was converted into a TensorFlow Lite for Microcontrollers (TFLM) model. This approach allows the DNN-based controller to operate within the real-time constraints of the TAB control loop while maintaining low memory and computational overhead [15].

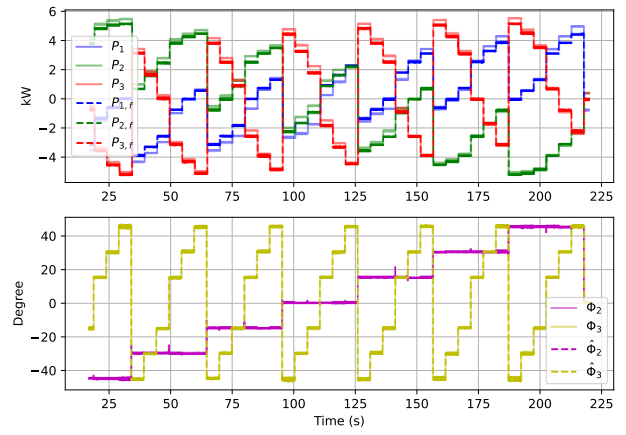


Fig. 6. Validation of the DNN under step changes of powers in each port.

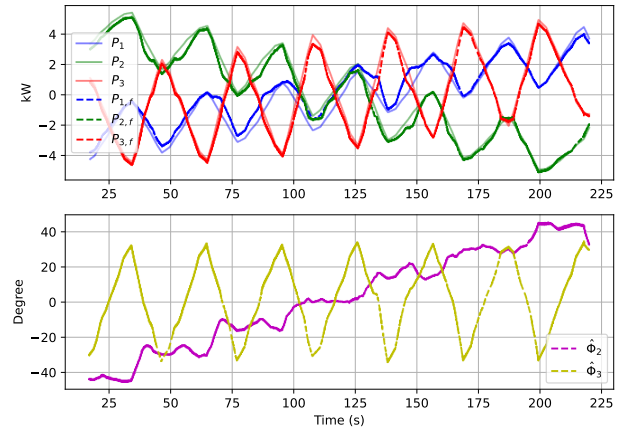


Fig. 7. Validation of the DNN under smooth changes of powers in each port.

IV. RESULTS AND DISCUSSION

The proposed control strategy is validated on a laboratory-scale experimental platform illustrated in Fig. 8. The testbed comprises a three-port MAB converter interfaced with three independent voltage sources. These sources are emulated using a Cinergia GE&EL-15 bidirectional power supply, allowing the behavior of three separate battery systems to be reproduced. The control algorithm is implemented on a low-cost STM32H745 microcontroller. The experimental parameters adopted in this study are summarized in Table I.

Fig. 9 shows system's response during startup, where a step power reference is applied to port 2. According to the power balance relation, the reference power of port 1 is equal in magnitude to that of port 2 but in the opposite direction. It can be observed that power is transferred from port 2 to port 1 smoothly and accurately, without ripples or overshoot. Subsequently, a positive step power reference is applied to port 3, and both ports 2 and 3 supply power to port 1, as shown in Fig. 10. The transient response remains smooth, and the coupling effects among the ports are effectively eliminated. This is clearly evidenced by the stable power of port 2 during

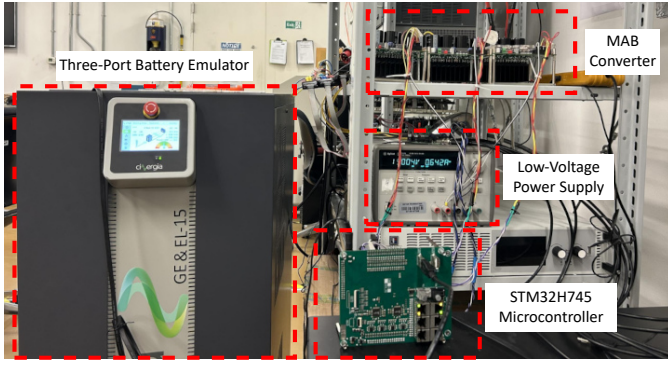


Fig. 8. Experimental three-port solid state transformer testbed.

TABLE I
SYSTEM'S PARAMETERS

Parameter	Symbol	Value
Port 1 voltage	V_1	600 V
Port 2 voltage	V_2	400 V
Port 3 voltage	V_3	400 V
Port side capacitance	C_y	1000 μ F
Leakage inductance	$L_{1,2,3,4}$	30 μ H
Port maximum power	P_{max}	6kW
Sampling time	T_s	10 μ s
Switching frequency	f_{sw}	100 kHz

the transient processes of ports 1 and 3.

Fig. 11 and Fig. 12 show the system response when the direction of power flow is suddenly reversed. It can be observed that each port smoothly and seamlessly regulates the power flow in a bidirectional manner, without ripples or overshoot. Finally, Fig. 13 illustrates the system behavior under voltage sag and voltage swell conditions. The input voltage of port 1 first experiences an 8% sag and then an 8% swell. In both scenarios, the power regulation performance is minimally affected, and the power of all ports remains stable, which demonstrates the robustness of the proposed control approach against external disturbances.

V. CONCLUSION

This paper presented a deep learning-based control strategy for multi-port solid-state transformers to achieve accurate, fast, and robust power flow regulation among multiple ports. A three-port SST architecture was developed, and the nonlinear relationship between inter-port power transfer and phase-shift modulation was systematically learned using a deep neural network trained on an extensive dataset spanning a wide range of operating conditions, including voltage variations and passive parameter uncertainties. By directly mapping power references to optimal phase-shift commands, the proposed controller eliminates the need for analytical power flow models and iterative optimization procedures, significantly reducing online computational complexity.

The results verified the effectiveness of the proposed approach under various operating scenarios. Smooth startup behavior, accurate steady-state power tracking, and effective decoupling among ports were demonstrated during step

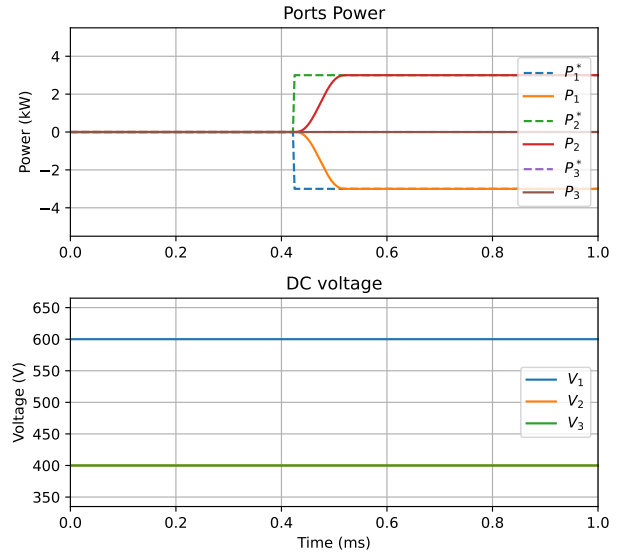


Fig. 9. Dynamic response of the system for stepping up port 2 power.

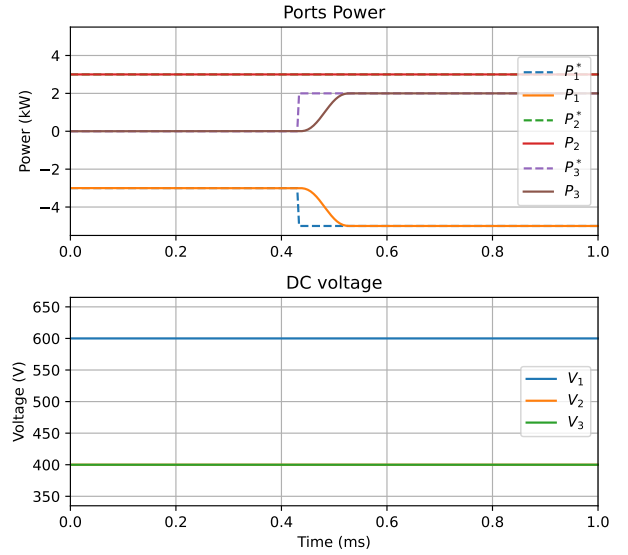


Fig. 10. Dynamic response of the system for stepping up port 3 power.

changes in power references. Furthermore, seamless bidirectional power flow regulation was achieved without overshoot or oscillations when the direction of power transfer was reversed. The controller also exhibited strong robustness against external disturbances, maintaining stable power regulation during voltage sag and swell conditions.

ACKNOWLEDGMENT

This publication was made possible by ARG01-0428-230023 from the Qatar Research, Development and Innovation (QRDI). The statements made herein are solely the responsibility of the authors.

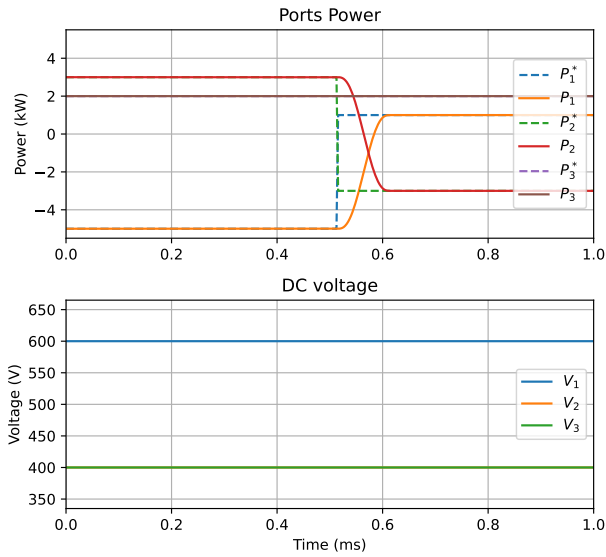


Fig. 11. Dynamic response of the system for port 2 bidirectional power flow.

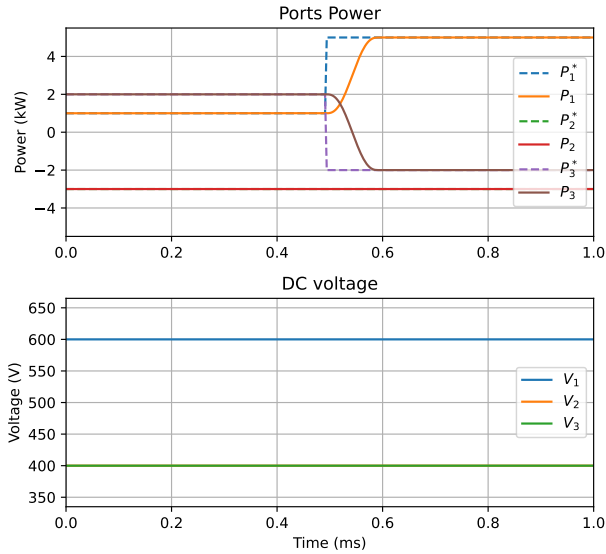


Fig. 12. Dynamic response of the system for port 3 bidirectional power flow.

REFERENCES

- [1] Y. Wu, C. Gao, Z. Lin, X. Wang, X. Jiang, Y. Yang, and G. Tang, "Multi-port arm shared dc-dc converter with decoupling control for multi-voltage dc grid," *IEEE Transactions on Power Electronics*, pp. 1–13, 2025.
- [2] A. Zade, S. Gurudiwan, and R. Zane, "98.3% efficient multiport system with multi-directional power flow to integrate battery storage/renewables with the grid for ev charging," *IEEE Transactions on Power Electronics*, pp. 1–25, 2025.
- [3] A. Sharida, A. Carreno, A. B. Bayindir, A. N. N. Alquennah, A. Karaki, S. Bayhan, H. Abu-Rub, and M. Malinowski, "Multiport smart transformers: A review of architectures, applications, and future trends," *IEEE Industrial Electronics Magazine*, 2025.
- [4] H. Wu, W. Lin, Y. Chen, H. Guo, M. Zhou, and R. Yang, "Study of the active disturbance rejection control for the decentralized multi-active bridge converter," *IEEE Transactions on Industry Applications*, pp. 1–12, 2025.

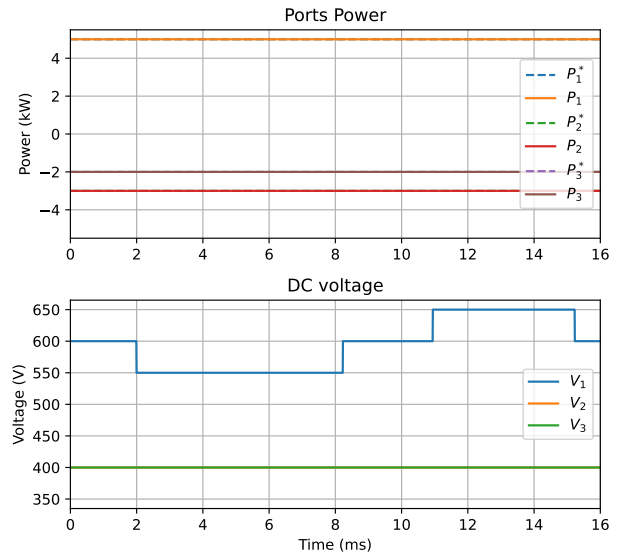


Fig. 13. Dynamic response of the system for port 1 voltage sag and swell.

- [5] P. Koohi, A. J. Watson, J. C. Clare, T. B. Soeiro, and P. W. Wheeler, "A survey on multi-active bridge dc-dc converters: Power flow decoupling techniques, applications, and challenges," *Energies*, vol. 16, no. 16, p. 5927, 2023.
- [6] L. Gong, J. Xu, J. Zhao, W. Li, T. Wang, Y. Peng, Y. Wang, and T. B. Soeiro, "A simplified all-zvs strategy for high-frequency triple active bridge converters with designed magnetizing inductance," *IEEE Transactions on Power Electronics*, vol. 38, no. 11, pp. 13 781–13 797, 2023.
- [7] S. Wang, F. Yu, and J. Qi, "Model predictive voltage control strategy for dual active bridge converters based on super-twisting integral sliding mode observer," *Electronics*, vol. 14, no. 8, p. 1496, 2025.
- [8] W. Wang, T. Liu, S. Cheng, C. Cheng, S. Niu, and A. Chen, "Feedback linearization-based load-current feedforward control strategy for fast dynamic response of hybrid modular dc transformers," *IEEE Transactions on Power Electronics*, 2025.
- [9] O. Babayomi, Y. Li, Z. Zhang, and K.-B. Park, "Advanced control of grid-connected microgrids: Challenges, advances, and trends," *IEEE Transactions on Power Electronics*, vol. 40, no. 6, pp. 7681–7708, 2025.
- [10] M. Liao, H. Li, P. Wang, T. Sen, Y. Chen, and M. Chen, "Machine learning methods for feedforward power flow control of multi-active-bridge converters," *IEEE Transactions on Power Electronics*, vol. 38, no. 2, pp. 1692–1707, 2023.
- [11] A. H. A. Adam, J. Chen, M. Xu, S. Kamel, G. I. Mustafa, Z. A. Zaki, and E. M. Ahmed, "Power decoupling enhancement of a triple active bridge converter with feedforward compensation based on model predictive control and fuzzy logic controller in dc microgrid systems," *IEEE Access*, vol. 12, pp. 140 310–140 328, 2024.
- [12] H. Ren, Y. Wang, H. Yu, B. Zhang, and Z. Chen, "Deep reinforcement learning-based power flow control for triple active bridge converter," in *2024 IEEE 10th International Power Electronics and Motion Control Conference (IPEMC2024-ECCE Asia)*, 2024, pp. 2107–2112.
- [13] A. Sharida, A. B. Bayindir, and S. Bayhan, "Inverse model predictive control for multi-port solid state transformer," in *2025 IEEE 19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG)*, 2025, pp. 1–6.
- [14] H. Yu, H. Zhang, Q. Zhang, Y. Wang, Z. Qin, Z. Chen, and P. Bauer, "A virtual impedance-based active damping control strategy for triple active bridge converter," *IEEE Transactions on Industrial Electronics*, vol. 71, no. 11, pp. 14 220–14 231, 2024.
- [15] N. F. Kamal, A. Sharida, S. Bayhan, H. Alnuweiri, and H. Abu-Rub, "Low-cost digital twin design for power electronics using deep neural networks," in *2024 4th International Conference on Smart Grid and Renewable Energy (SGRE)*, 2024, pp. 1–6.